







The sciences have identified restrictions on

**when (src) {**  **src.balance += 10; }**

**when (dst) {**  **dst.balance += 10; }**

3

6

**when(dst) {**  **print(dst.balance) }**

**when(src) {**  **print(src.balance) }**

src=0, dst=0





src=0, dst=10



src=10, dst=0



src=10, dst=10



Order

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Permitted

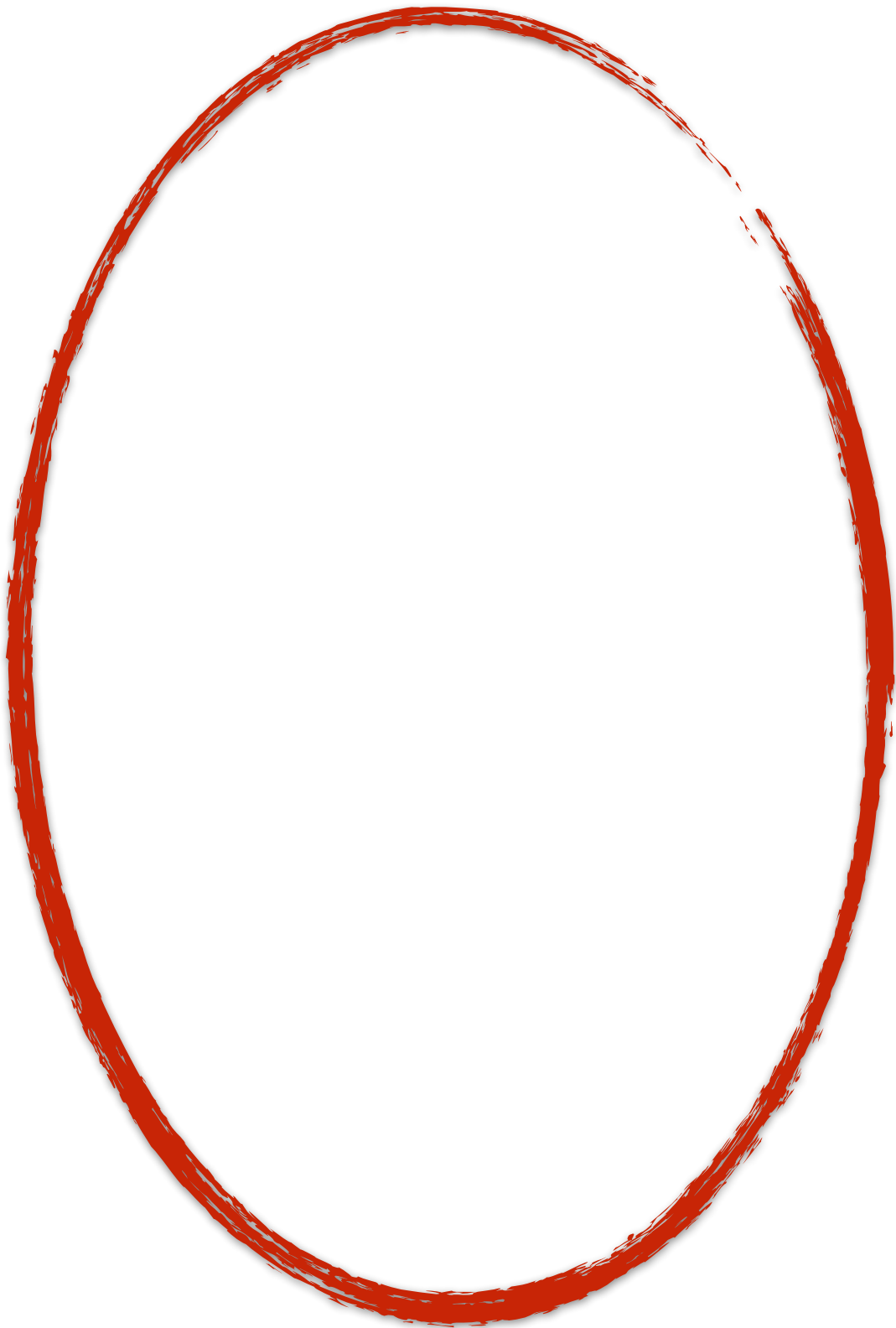




updatepath(src then dst)

Diagnostic path (dst then src)





# The semantics have incidental restrictions on executions

Update path (src then dst)

**when**(src) { A src.balance += 10; }

**when**(dst) { B dst.balance += 10; }

Diagnostic path (dst then src)

**when**(dst) { C print(dst.balance) }

**when**(src) { D print(src.balance) }

|           | src=0, dst=0                                                                                                                                                                                                                                                                                                                                                                                                            | src=0, dst=10                                                                                                                                                                                                                                                                                                                                                                                                           | src=10, dst=0                                                                                                                                                                                                                                                                                                                                                                                                           | src=10, dst=10                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Order     | <span style="background-color: #4682b4; color: white; border-radius: 50%; padding: 2px 5px;">C</span> <span style="background-color: #add8e6; color: black; border-radius: 50%; padding: 2px 5px;">B</span> <span style="background-color: #008000; color: white; border-radius: 50%; padding: 2px 5px;">D</span> <span style="background-color: #4b4b9b; color: white; border-radius: 50%; padding: 2px 5px;">A</span> | <span style="background-color: #add8e6; color: black; border-radius: 50%; padding: 2px 5px;">B</span> <span style="background-color: #4682b4; color: white; border-radius: 50%; padding: 2px 5px;">C</span> <span style="background-color: #008000; color: white; border-radius: 50%; padding: 2px 5px;">D</span> <span style="background-color: #4b4b9b; color: white; border-radius: 50%; padding: 2px 5px;">A</span> | <span style="background-color: #4b4b9b; color: white; border-radius: 50%; padding: 2px 5px;">A</span> <span style="background-color: #4682b4; color: white; border-radius: 50%; padding: 2px 5px;">C</span> <span style="background-color: #add8e6; color: black; border-radius: 50%; padding: 2px 5px;">B</span> <span style="background-color: #008000; color: white; border-radius: 50%; padding: 2px 5px;">D</span> | <span style="background-color: #4b4b9b; color: white; border-radius: 50%; padding: 2px 5px;">A</span> <span style="background-color: #add8e6; color: black; border-radius: 50%; padding: 2px 5px;">B</span> <span style="background-color: #4682b4; color: white; border-radius: 50%; padding: 2px 5px;">C</span> <span style="background-color: #008000; color: white; border-radius: 50%; padding: 2px 5px;">D</span> |
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| Permitted | ✓                                                                                                                                                                                                                                                                                                                                                                                                                       | ⚠                                                                                                                                                                                                                                                                                                                                                                                                                       | ✓                                                                                                                                                                                                                                                                                                                                                                                                                       | ✓                                                                                                                                                                                                                                                                                                                                                                                                                       |

# There are restrictions we want to keep

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```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    }  
  }  
}
```

A nested behaviour is spawned synchronously,  
but can run independently of it's parent